

Class 15: Symmetric Groups

Goals

1. Understand permutations as functions.
2. Understand array notation and cycle notation.
3. Calculate the number of permutations in S_n .

Let S_3 be the set of all functions with domain $\{1,2,3\}$ and range $\{1,2,3\}$. List all of the distinct functions in S_3 .

χ	$\xi(x)$	$\alpha(x)$	$\beta(x)$	$\gamma(x)$	$\delta(x)$	$\varphi(x)$
1	1	1	2	2	3	3
2	2	3	1	3	1	2
3	3	2	3	1	2	1

Each function can be thought of as an arrangement of the numbers 1, 2, 3. These functions are one-to-one and onto from the set $\{1,2,3\}$ to itself. Such functions are called _____ on the set $\{1,2,3\}$.

Big idea: The set of all permutations on the set $\{1,2,\dots, n\}$ forms a group.

under the operation composition of function.

The group is denoted by S_n and called the symmetric group on n objects.

Array notation: Instead of writing a permutation in S_n using a vertical table of x-values and function values, we can use a horizontal two-dimensional array.

Examples in S_3

$$\varepsilon = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} \quad \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} \quad \beta = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\gamma =$$

$$\delta =$$

$$\varphi =$$

Explore the group operation

$$\varepsilon \circ \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} = \alpha$$

$$\varepsilon \circ \beta = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} = \beta$$

$$\beta \circ \varepsilon = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} = \beta$$

$$\alpha \circ \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} = \varepsilon$$

$$\alpha \circ \beta = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} = \delta$$

$$\beta \circ \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \gamma$$

Make a complete Cayley table for S_3 .

$\circ$	ε	α	β	γ	δ	φ
ε						
α		$\alpha \circ \beta$				
β		$\beta \circ \alpha$				
γ						
δ						
φ						

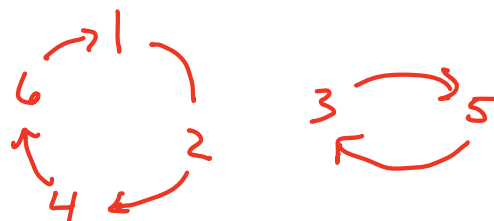
The number of elements in a group is called the order of the group.

How many permutations are there in S_3 , S_4 , or S_n ?

$$|S_3| = 3! = 6 \quad |S_4| = 4! = 24 \quad |S_n| = n!$$

Cycle notation is a more efficient way to represent permutations for larger values of n .

Example: $\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 5 & 6 & 3 & 1 \end{bmatrix}$



Write $\alpha = (1\ 2\ 4\ 6)(3\ 5)$

$$\alpha^2 = (1\ 4)(2\ 6) \quad \alpha^3 = (1\ 6\ 4\ 2)(3\ 5)$$

$$\alpha^4 = (1)$$

Let $\alpha = (1246)(35)$ and $\beta = (135)(246)$.

$$\begin{aligned} \alpha \circ \beta &= (1246)(3\ 5)(135)(246) \\ &= (1\ 5\ 2\ 6\ 4)(3) \end{aligned}$$

Standard cycle form

Cycles are disjoint *no number appears in more than one cycle*

Smallest numbers first. *The leading number in each cycle is the smallest number in that cycle.*

Leading numbers in increasing order *The leading number in each cycle is larger than any previous leading number.*

Don't list fixed numbers *If a cycle only has one number, don't list it -*

Exception for the identity permutation $\varepsilon = (1)$

Practice: Write each permutation in standard cycle notation form.

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 3 & 5 & 7 & 2 & 4 & 6 \end{bmatrix}$$

$$\alpha = (2\ 3\ 5)(4\ 7\ 6)$$

$$\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 6 & 5 & 4 & 3 & 2 & 1 \end{bmatrix}$$

$$\beta = (1\ 7)(2\ 6)(3\ 5)$$

$$\gamma = \alpha \circ \beta = (1\ 6\ 3\ 2\ 4\ 7)$$

$$\gamma^{-1} = (1\ 7\ 4\ 2\ 3\ 6)$$

$$\gamma^2 = (1\ 3\ 4)(2\ 7\ 6)$$

List all distinct powers of γ